ModelIt!

Student Activity Pack

How Can We Protect Against Natural Hazards?

Engineering Solutions for Dangerous Events

Grade 3 | NGSS 3-ESS3-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What natural hazards happen where you live — floods, earthquakes, tornadoes, hurricanes, or wildfires?**

2. **How do people protect their homes from dangerous weather events?**

3. **Draw a building that you think could survive a flood, and explain what makes it strong.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Hazard Strength | Protection Design | Damage Reduction

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Weak Storm, No Protection

Hazard Strength low, Protection Design none

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Strong Storm, Strong Protection

Hazard Strength high, Protection Design maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Strong Storm, Weak Protection

Hazard Strength high, Protection Design low

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Warning Time — How much advance notice people have before the hazard hits — more warning time allows better preparation

Building Materials — What the structure is made of — some materials resist water, wind, or shaking better than others

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can people build things that protect them from dangerous natural events like floods, tornadoes, and earthquakes? Use evidence from your simulation to support your answers.

1 Why can not engineers build something that protects against ALL natural hazards perfectly?

2 How did your model help you understand why some buildings survive storms while others do not?

8 STEM Challenge: Design a Flood-Proof Building

ModelIt!

YOUR ENGINEERING MISSION

Design and build a model structure that can survive a flood by keeping the inside dry when water rises around it.

THE SCENARIO

A town near a river floods every spring. The mayor has hired your engineering team to design a building that stays dry even when floodwater rises two feet high. Use what you learned about hazard protection to create your design!

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE FOR YOUR BUILDING — WATERPROOF OR WATER-RESISTANT MATERIALS?

2. HOW WILL YOU SEAL THE BOTTOM AND SIDES TO PREVENT WATER FROM SEEPING IN?

3. WILL YOU RAISE THE BUILDING ON STILTS OR PILLARS ABOVE THE FLOOD LEVEL?

4. HOW WILL YOU TEST YOUR DESIGN — PLACING IT IN A TRAY OF WATER AND SLOWLY RAISING THE LEVEL?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Civil Engineers design bridges, buildings, dams, and other structures that keep people safe from natural hazards. They earn $60,000-$110,000/year.